Simulation-Based Evaluation of Geometric and Particle Filters for Multi-Camera UAV Tracking

Abstract
Multi-camera vision-based tracking systems provide a promising alternative to radar in urban environments where line-of-sight constraints degrade traditional sensing. While particle filters (PF) have demonstrated robustness in such systems, a systematic comparison with geometric filtering methods remains limited. This work presents a simulation-based evaluation of PF, multiplicative extended Kalman filter (MEKF), invariant extended Kalman filter (InEKF), and feedback particle filter (FPF) within a unified multi-camera tracking framework. A consistent simulation environment is developed using a calibrated camera network and perspective projection model with noisy image measurements. Performance is evaluated across scenarios including nominal tracking, poor initialization, maneuvering targets, and measurement ambiguity. Preliminary results indicate that PF exhibits strong robustness under ambiguity, while geometric filters offer computational efficiency and stability in moderate noise regimes. FPF demonstrates potential advantages but remains sensitive to gain approximation. This study highlights key trade-offs between filtering approaches and establishes a foundation for future experimental validation.

1. Introduction
Urban airspace monitoring for unmanned aerial systems (UAS) presents significant challenges due to occlusions and limited line-of-sight in dense environments. Camera networks offer a scalable alternative for tracking aerial targets by combining multi-view measurements to estimate 3D trajectories. Prior work has shown that particle filters can effectively fuse these measurements, but comparisons with alternative filtering approaches remain limited. This work develops a unified framework to evaluate multiple filtering techniques under identical simulation conditions and analyze their performance in realistic tracking scenarios.

2. Methodology
The target state is modeled using a constant-velocity system with position and velocity components. Measurements are generated using a perspective projection model mapping 3D positions to 2D image coordinates across multiple cameras. Noise is introduced in pixel measurements to simulate realistic detection uncertainty. The filters evaluated include: (1) Particle Filter (PF), (2) Multiplicative Extended Kalman Filter (MEKF), (3) Invariant Extended Kalman Filter (InEKF), and (4) Feedback Particle Filter (FPF).

3. Simulation Setup
A simulated environment is constructed with four calibrated cameras positioned around a tracking volume. Targets follow predefined trajectories including linear, maneuvering, and crossing paths. Gaussian noise is added to image measurements, and scenarios with poor initialization and measurement ambiguity are included to stress test filter performance.

4. Preliminary Results
Preliminary simulations show that PF performs robustly in nonlinear and ambiguous scenarios due to its multi-hypothesis representation. MEKF and InEKF provide efficient and stable estimates under moderate noise conditions, with InEKF demonstrating improved consistency in nonlinear regimes. FPF exhibits promising behavior by combining particle flexibility with feedback structure, though performance depends on gain approximation quality.

5. Future Work
Future work will extend the simulation study with Monte Carlo evaluations, improved FPF gain approximation, and integration with real-world camera data. Additional metrics including computational

efficiency and robustness to missed detections will be analyzed.